



Mayer Topology in TDA

October 22, 2025

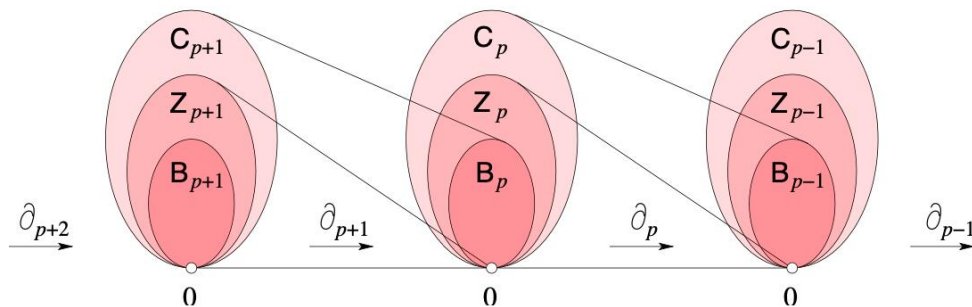
Talk by Dilan Karaguler(karagule@msu.edu)

Fall 2025 MTH994:

Mathematics-Assisted AI and AI-inspired Mathematics

Guo-Wei Wei (weig@msu.edu)

Standard homology



- Chain Complex

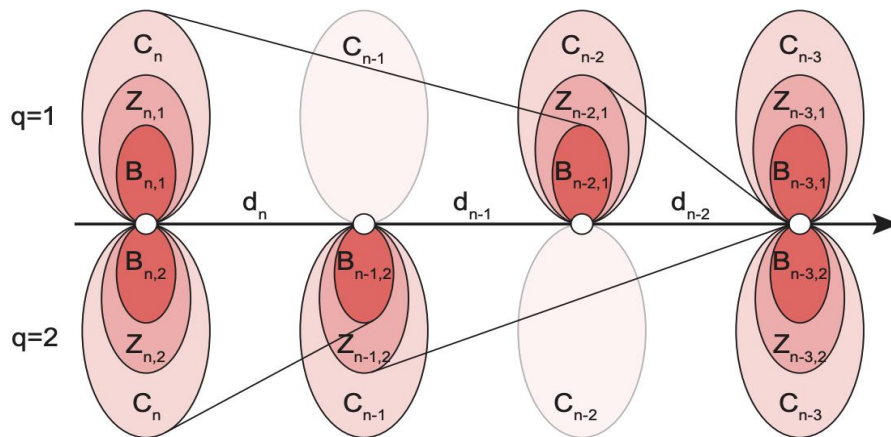
- $C_*: \cdots C_3 \xrightarrow{d_3} C_2 \xrightarrow{d_2} C_1 \xrightarrow{d_1} C_0 \rightarrow 0$
- Chain groups: $C_n = \text{span}_{\mathbb{K}}\{n - \text{simplices}\}$
- Chain maps: $d_n: C_n \rightarrow C_{n-1}$ where $d^2 = 0$

- Homology

- $$H_n(C_*) = Z_n(C_*)/B_n(C_*)$$

$$= \ker d_n / \text{im } d_{n+1}$$

Mayer homology



■ N-Chain Complex

- $C_*: \cdots C_3 \xrightarrow{d_3} C_2 \xrightarrow{d_2} C_1 \xrightarrow{d_1} C_0 \rightarrow 0$
- $C_n = \text{span}_{\mathbb{K}}\{n - \text{simplices}\}$
- $d_n: C_n \rightarrow C_{n-1}$ where $d^N = 0$

■ Homology for any $1 \leq q \leq N - 1$

- $H_n^{N,q}(C_*) = Z_n^{N,q}(C_*) / B_n^{N,q}(C_*)$
 $= \ker d^q / \text{im } d^{N-q}$
- $H_n^N(C_*) = \bigoplus_q H_n^{N,q}(C_*)$

Mayer homology history-timeline

1. Mayer, A new homology theory, 1942
 - Introduced the notion of N-complex as $d^N = 0$
2. Spainer, The Mayer Homology Theory, 1949
 - If coefficient field has characteristic p , Mayer homology will coincide with standard homology.
3. Dunois-Violette, $d^N = 0$, 1990
 - Formalized Mayer's construction in the language of N-complexes and generalized homology
4. Kapranov, On the q th analog of homological algebra, 1996
 - Construction of simplicial complex by using N-th primitive root of unity
5. Tikaradze, Homological construction on N-complexes, 2001
 - Euler Poincare formula that is generalized Euler char for Mayer groups.

Mayer vs Standard homology

Let $\partial_i^n([v_0, \dots, v_n]) = [v_0, \dots, \widehat{v_i}, \dots, v_n]$ be the face map

- Standard simplicial chain map

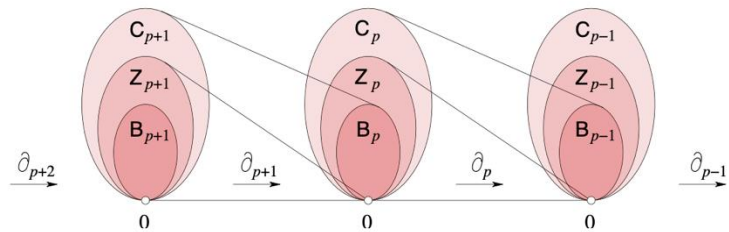
- $d_n = \sum_{i=0}^n (-1)^i \partial_i^n$
- $d^2 = 0$

- N-invariant simplicial chain map

- $d_n = \sum_{i=0}^n (\xi)^i [v_0, \dots, \widehat{v_i}, \dots, v_n]$ where ξ is the Nth primitive root of unity ($\xi = e^{\frac{2\pi i}{N}}$)
- $d^N = (\prod_{k=1}^N (1 + \xi + \dots + \xi^{k-1})) \sum_{j_1 < \dots < j_N} \xi^{j_1 + \dots + j_N} \partial_{j_1} \dots \partial_{j_N}$
- $1 + \xi + \dots + \xi^{N-1} = 0$ implies $d^N = 0$

Mayer Homology for N=2

$$q \in \{1\} \text{ and } \xi = -1, d_n = \sum_{i=0}^n (\xi)^i [v_0, \dots, \widehat{v}_i, \dots, v_n] = \sum_{i=0}^n (-1)^i [v_0, \dots, \widehat{v}_i, \dots, v_n]$$



$$C_*: \cdots C_3 \xrightarrow{d} C_2 \xrightarrow{d} C_1 \xrightarrow{d} C_0 \rightarrow 0$$

- $H_n^{2,1}(C_*) = Z_n^{2,1}(C_*)/B_n^{2,1}(C_*) = \ker d^1 / \text{im} d^1 = H_n(C_*).$
- Mayer homology for N=2 is the same with standard homology

Mayer Homology for N=3

$$q \in \{1,2\} \text{ and } \xi = -\frac{1}{2} + i \frac{\sqrt{3}}{2}, d_n = \sum_{i=0}^n (\xi)^i [v_0, \dots, \widehat{v_i}, \dots, v_n]$$

For q=1

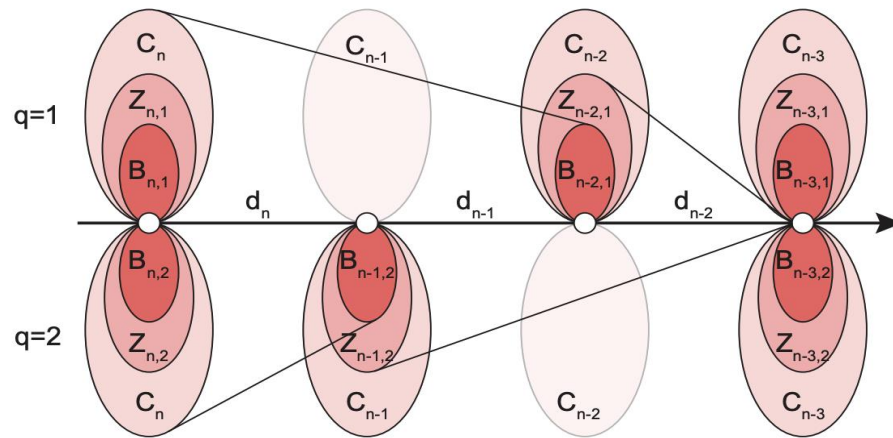
$$C_*: \dots C_{n+2} \xrightarrow{d^2} C_n \xrightarrow{d} C_{n-1} \xrightarrow{d^2} \dots$$

- $H_n^{3,1}(C_*) = Z_n^{3,1}(C_*)/B_n^{3,1}(C_*)$
 $= \ker d^1 / \text{imd}^2$

For q=2

$$C_*: \dots C_{n+1} \xrightarrow{d} C_n \xrightarrow{d^2} C_{n-2} \xrightarrow{d} \dots$$

- $H_n^{3,2}(C_*) = Z_n^{3,2}(C_*)/B_n^{3,2}(C_*)$
 $= \ker d^2 / \text{imd}^1$



Mayer homology for general N

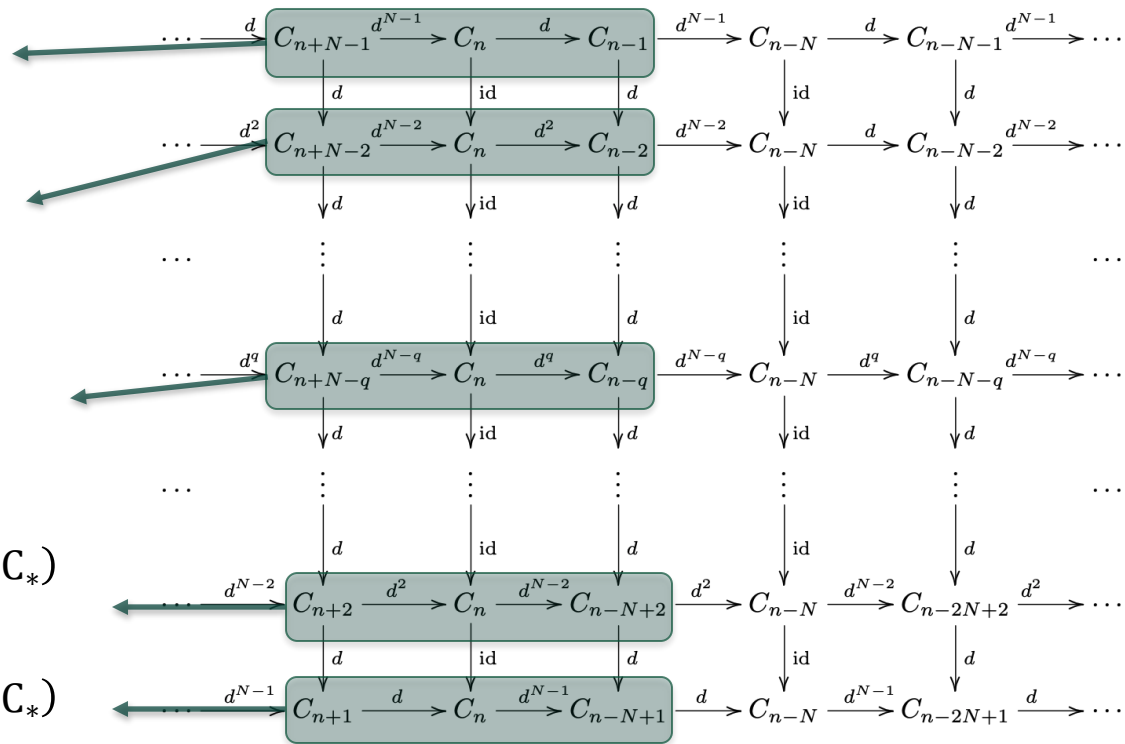
$$H_n^{N,1}(C_*) = Z_n^{N,1}(C_*)/B_n^{N,1}(C_*) \\ = \ker d^1 / \text{imd}^{N-1}$$

$$H_n^{N,2}(C_*) = Z_n^{N,2}(C_*)/B_n^{N,2}(C_*) \\ = \ker d^2 / \text{imd}^{N-2}$$

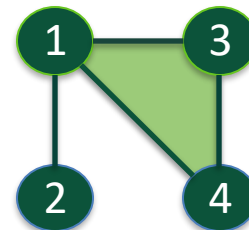
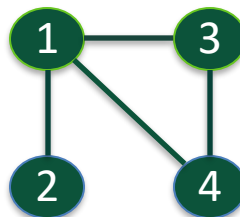
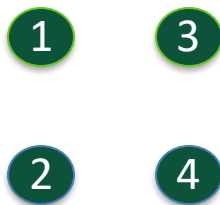
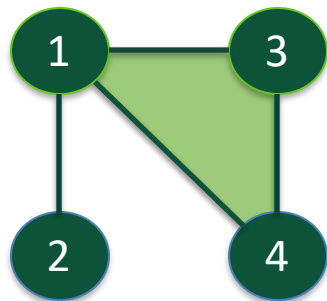
$$H_n^{N,q}(C_*) = Z_n^{N,q}(C_*)/B_n^{N,q}(C_*) \\ = \ker d^q / \text{imd}^{N-q}$$

$$H_n^{N,N-2}(C_*) = Z_n^{N,N-2}(C_*)/B_n^{N,N-2}(C_*) \\ = \ker d^{N-2} / \text{imd}^2$$

$$H_n^{N,N-1}(C_*) = Z_n^{N,N-1}(C_*)/B_n^{N,N-1}(C_*) \\ = \ker d^{N-1} / \text{imd}^1$$



Example



$$C_0 = \langle v_1, v_2, v_3, v_4 \rangle = \mathbb{C}^4$$

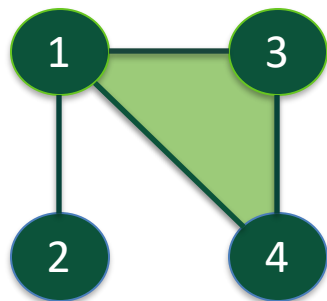
$$C_1 = \langle v_{12}, v_{13}, v_{14}, v_{34} \rangle = \mathbb{C}^4$$

$$C_2 = \langle v_{134} \rangle = \mathbb{C}$$

$$d_1 = \begin{matrix} & v_{12} & v_{13} & v_{14} & v_{34} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{matrix} & \begin{pmatrix} \xi & \xi & 0 & \xi \\ 1 & 0 & 0 & 0 \\ 0 & 1 & \xi & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} \end{matrix} = \begin{matrix} & v_{12} & v_{13} & v_{14} & v_{34} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{matrix} & \begin{pmatrix} -1 & -1 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} \end{matrix} \quad d_2 = \begin{matrix} & v_{134} \\ \begin{matrix} v_{12} \\ v_{13} \\ v_{14} \\ v_{34} \end{matrix} & \begin{pmatrix} 0 \\ \xi^2 \\ \xi \\ 1 \end{pmatrix} \end{matrix} = \begin{matrix} & v_{134} \\ \begin{matrix} v_{12} \\ v_{13} \\ v_{14} \\ v_{34} \end{matrix} & \begin{pmatrix} 0 \\ 1 \\ -1 \\ 1 \end{pmatrix} \end{matrix}$$

$$H_0^{2,1} = \mathbb{C} \quad H_1^{2,1} = 0 \quad H_2^{2,1} = 0. \text{ (Lecture 2, Jones)}$$

Example N=3, q=1

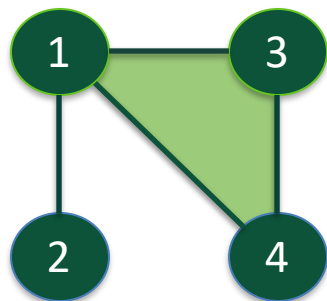


$$C_*: \cdots 0 \xrightarrow{d_3} C_2 \xrightarrow{d_2} C_1 \xrightarrow{d_1} C_0 \xrightarrow{d_0} 0$$

$$d_1 = \begin{matrix} & v_{12} & v_{13} & v_{14} & v_{34} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{matrix} & \begin{pmatrix} \xi & \xi & \xi & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & \xi \\ 0 & 0 & 1 & 1 \end{pmatrix} \end{matrix} \quad d_2 = \begin{matrix} & v_{134} \\ \begin{matrix} v_{12} \\ v_{13} \\ v_{14} \\ v_{34} \end{matrix} & \begin{pmatrix} 0 \\ \xi^2 \\ \xi \\ 1 \end{pmatrix} \end{matrix} \quad d_1 d_2 = \begin{matrix} & v_{134} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{matrix} & \begin{pmatrix} \xi^3 + \xi^2 \\ 0 \\ \xi^2 + \xi \\ 1 + \xi \end{pmatrix} \end{matrix}$$

- $H_0^{3,1} = Z_0^{3,1}/B_0^{3,1} = C_0/\text{im}(d_1 d_2)$
 $\leftarrow C_*: 0 \rightarrow C_2 \xrightarrow{d_1 d_2} C_0 \xrightarrow{d_0} 0$
 $= \langle v_1, v_2, v_3, v_4 \rangle / \langle (1 + \xi^2)v_1 + (\xi^2 + \xi)v_3 + (1 + \xi)v_4 \rangle$
 $= \langle v_1, v_2, v_3, v_4 \rangle / \langle -\xi v_1 - v_3 - \xi^2 v_4 \rangle = \mathbb{C}^3$
- $H_1^{3,1} = Z_1^{3,1}/B_1^{3,1} = \ker d_1 / \text{im}(d_2 d_3) = 0$ since d_1 has rank 4.
 $\leftarrow C_*: 0 \xrightarrow{d_2 d_3} C_1 \xrightarrow{d_1} C_0 \rightarrow 0$
- $H_2^{3,1} = Z_2^{3,1}/B_2^{3,1} = \ker d_2 / \text{im}(d_3 d_4) = 0$
 $\leftarrow C_*: 0 \xrightarrow{d_3 d_4} C_2 \xrightarrow{d_2} C_1 \rightarrow 0$

Example N=3, q=2



$$d_1 = \begin{matrix} & v_{12} & v_{13} & v_{14} & v_{34} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{matrix} & \begin{pmatrix} \xi & \xi & \xi & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & \xi \\ 0 & 0 & 1 & 1 \end{pmatrix} \end{matrix}$$

$$d_2 = \begin{matrix} & v_{134} \\ \begin{matrix} v_{12} \\ v_{13} \\ v_{14} \\ v_{34} \end{matrix} & \begin{pmatrix} 0 \\ \xi^2 \\ \xi \\ 1 \end{pmatrix} \end{matrix}$$

$$d_1 d_2 = \begin{matrix} & v_{134} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{matrix} & \begin{pmatrix} \xi^3 + \xi^2 \\ 0 \\ \xi^2 + \xi \\ 1 + \xi \end{pmatrix} \end{matrix}$$

- $H_0^{3,2} = Z_0^{3,2}/B_0^{3,2} = C_0/\text{im}(d_1)$

$$= \langle v_1, v_2, v_3, v_4 \rangle / \langle v_2 + \xi v_1, v_3 + \xi v_1, v_4 + \xi v_1, v_4 + \xi v_3 \rangle = 0$$

since d_1 has rank 4.

- $H_1^{3,2} = Z_1^{3,2}/B_1^{3,2} = C_1/\text{im}(d_2)$

$$= \langle v_{12}, v_{13}, v_{14}, v_{34} \rangle / \langle v_{34} + \xi v_{14} + \xi^2 v_{13} \rangle = \mathbb{C}^3$$

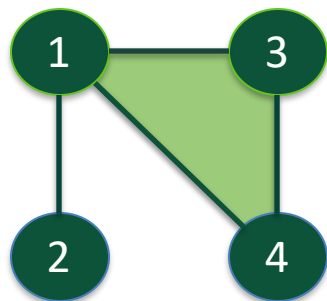
- $H_2^{3,2} = Z_2^{3,2}/B_2^{3,2} = \ker(d_1 d_2)/\text{im}(d_3) = 0$

$$C_*: 0 \rightarrow C_1 \xrightarrow{d_1} C_0 \rightarrow 0$$

$$C_*: 0 \rightarrow C_2 \xrightarrow{d_2} C_1 \rightarrow 0$$

$$C_*: 0 \xrightarrow{d_3} C_2 \xrightarrow{d_1 d_2} C_0 \xrightarrow{d_0} 0$$

Example N=3, q=2



$$d_1 = \begin{matrix} & v_{12} & v_{13} & v_{14} & v_{34} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{matrix} & \begin{pmatrix} \xi & \xi & \xi & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & \xi \\ 0 & 0 & 1 & 1 \end{pmatrix} \end{matrix} \quad d_2 = \begin{matrix} & v_{134} \\ \begin{matrix} v_{12} \\ v_{13} \\ v_{14} \\ v_{34} \end{matrix} & \begin{pmatrix} 0 \\ \xi^2 \\ \xi \\ 1 \end{pmatrix} \end{matrix} \quad d_1 d_2 = \begin{matrix} & v_{134} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{matrix} & \begin{pmatrix} \xi^3 + \xi^2 \\ 0 \\ \xi^2 + \xi \\ 1 + \xi \end{pmatrix} \end{matrix}$$

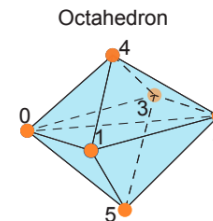
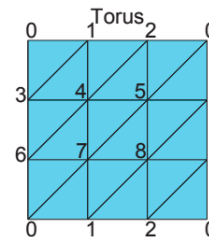
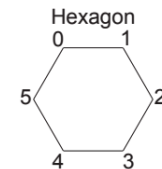
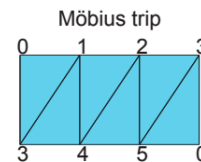
- $H_0^{2,1} = \mathbb{C} \quad H_1^{2,1} = 0 \quad H_2^{2,2} = 0$
- $H_0^{3,1} = \mathbb{C}^3 \quad H_1^{3,1} = 0 \quad H_2^{3,1} = 0$
- $H_0^{3,2} = 0 \quad H_1^{3,2} = \mathbb{C}^3 \quad H_2^{3,2} = 0$

Mayer Betti Number

Betti numbers are the dimension of the homology groups on the corresponding dimension

$$\beta_n^{N,q} = \dim H_n^{N,q}$$

simplicial complexes	$\beta_{0,1}$	$\beta_{1,1}$	$\beta_{2,1}$	$\beta_{0,2}$	$\beta_{1,2}$	$\beta_{2,2}$
$\Delta[3]$	1	1	0	0	2	0
$\partial\Delta[3]$	1	2	0	0	2	1
Hexagon	6	0	0	0	6	0
Möbius trip	1	6	0	0	6	1
Torus	1	18	0	0	9	10
Octahedron	1	3	1	0	2	3



Euler Char and Poincare Polynomial

1. Euler char :

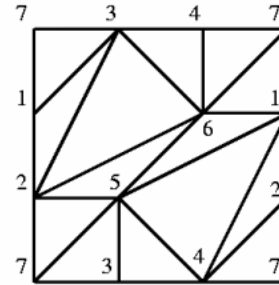
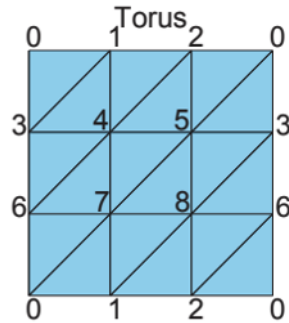
- $\chi = \sum_n (-1)^n \dim(C_n) = \sum_n (-1)^n \dim(H_n)$
- 2-exact sequence :
- Vanishes at 2-exact sequences ($\ker d = \operatorname{Im} d$)

2. Poincare polynomial :

- $P(\xi) = \sum_n (\xi)^n \dim(C_n) = \frac{1}{1-\xi^q} \sum_n (\xi)^n (\dim(H_n^{N,q}) - \dim(H_n^{N,N-q}))$, 2001 Tikaradze
- N-exact sequence : $\ker d^q = \operatorname{Im} d^{N-q}$ for all q
- Vanishes at N-exact sequence, 1996 Kapranov
- When N=2, it coincide with Euler char

Mayer Betti Number is not an invariant

$$\begin{aligned}\beta_0^{3,1} &= 1 \\ \beta_1^{3,1} &= 18 \\ \beta_2^{3,1} &= 0 \\ \beta_0^{3,2} &= 0 \\ \beta_1^{3,2} &= 9 \\ \beta_2^{3,2} &= 10\end{aligned}$$



$$\begin{aligned}\beta_0^{3,1} &= 0 \\ \beta_1^{3,1} &= 14 \\ \beta_2^{3,1} &= 0 \\ \beta_0^{3,2} &= 0 \\ \beta_1^{3,2} &= 7 \\ \beta_2^{3,2} &= 7\end{aligned}$$

Where will Mayer Betti Numbers be a good fit?

1. Example of two different size of loop
 - **Functional networks**
 - In **biological or gene networks**, a **small loop** (triangle) often indicates **strong local feedback** or motif (like a regulatory circuit).
 - A **large loop** may indicate a **long-range interaction pathway**, or a cycle involving many nodes with weaker direct influence.
 - **Social networks**
 - Small loops = tightly knit communities (triangles → strong clustering).
 - Large loops = loose associations or bridges connecting communities.
2. **Network comparison / classification** - Loop-size sensitivity allows you to detect:
 - Local motifs detection

How to compute Mayer Betti Numbers?

Let M_n^q be the representation matrix of $d_{n-q+1} \cdots d_{n-1} d_n: C_n \rightarrow C_{n-q}$

$$\beta_n^{N,q} = \dim C_n - \text{rank}(M_n^q) - \text{rank}(M_{n+N-q}^{N-q})$$

The fundamental exact sequences:

$$0 \rightarrow Z_n^{N,q} \rightarrow C_n \xrightarrow{d^q} B_{n-q}^{N,N-q} \rightarrow 0$$

$$0 \rightarrow B_n^{N,q} \rightarrow Z_n^{N,q} \rightarrow H_n^{N,q} \rightarrow 0$$

Coefficients from a field $\Rightarrow$ split ie $C_n = Z_n^{N,q} \oplus B_{n-q}^{N,N-q}$ and $Z_n^{N,q} = H_n^{N,q} \oplus B_n^{N,q}$

$$\dim(C_n) = \dim(Z_n^{N,q}) + \dim(B_{n-q}^{N,N-q}) = \dim(H_n^{N,q}) + \dim(B_n^{N,q}) + \dim(B_{n-q}^{N,N-q})$$

$$\dim(C_n) = \beta_n^{N,q} + \text{rank}(M_{n+N-q}^{N-q}) + \text{rank}(M_n^q)$$

Persistent Mayer Homology

- Mayer homology $H_*^{N,q}$ is a functor from Nchain to Vec for each $1 \leq q \leq N - 1$

Let $C_*(K_a)$ be a N-chain complex for each a so that

$$\text{Filtration: } C_*(K_{a_1}) \subset C_*(K_{a_2}) \subset \cdots \subset C_*(K_{a_m})$$

$$\text{Induced maps: } H_*^{N,q}(C_*(K_{a_1})) \subset H_*^{N,q}(C_*(K_{a_2})) \subset \cdots \subset H_*^{N,q}(C_*(K_{a_m}))$$

- Persistent classes (a,b)-persistent Mayer homology

$$H_{n,(a,b)}^{N,q} := \text{im}\{H_n^{N,q}(C_*(K_a)) \rightarrow H_n^{N,q}(C_*(K_b))\}$$

- The rank of $H_{n,(a,b)}^{N,q}$ is the (a,b)-persistent Betti numbers.

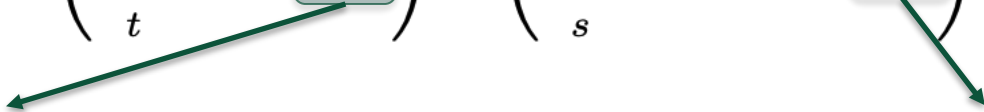
How to visualize the elements in the persistent module

$$H_q = \bigoplus_{i=1}^{\infty} H_*^{N,q}(K_{a_i})$$

$$x: H_q \rightarrow H_q \text{ so that } x: H_*^{N,q}(K_{a_i}) \rightarrow H_*^{N,q}(K_{a_{i+1}})$$

H_q is $\mathbb{C}[x]$ -module where $(f(x) \cdot \alpha)$ shifts the α with x component

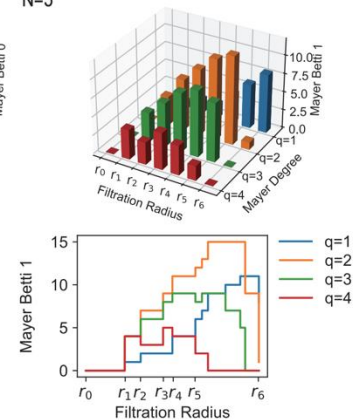
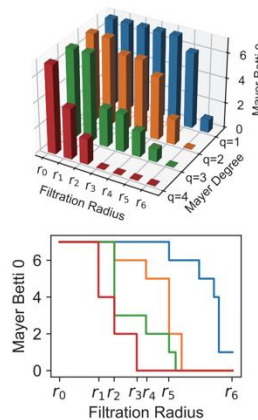
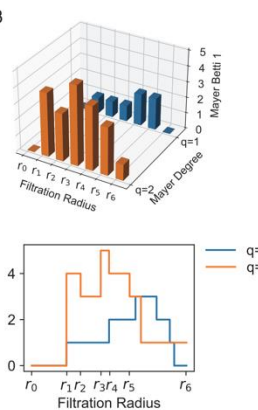
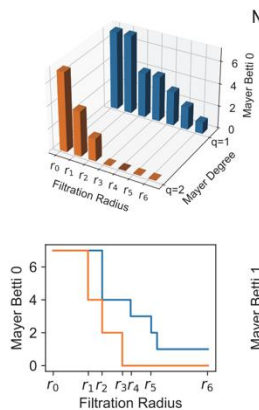
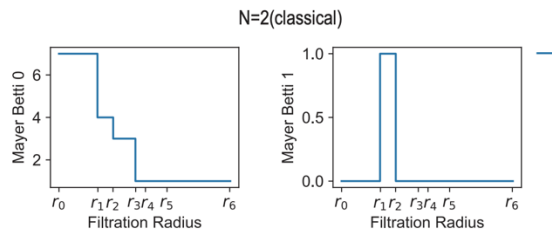
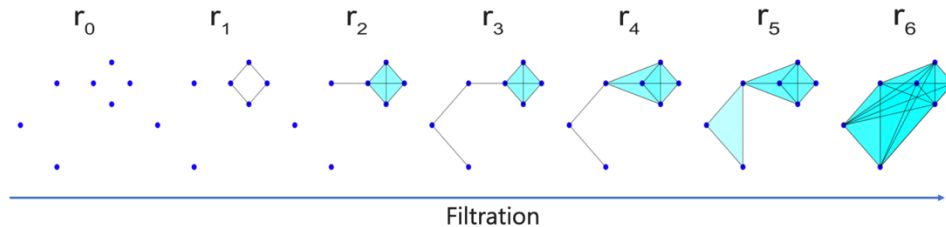
$\mathbb{C}[x]$ -modules will have decomposition since $\mathbb{C}[x]$ is PID

$$\mathbf{H}_q \cong \left(\bigoplus_t \mathbb{C}[x] \cdot \alpha_{b_t} \right) \oplus \left(\bigoplus_s \mathbb{C}[x]/x^{c_s} \cdot \beta_{b_s} \right)$$


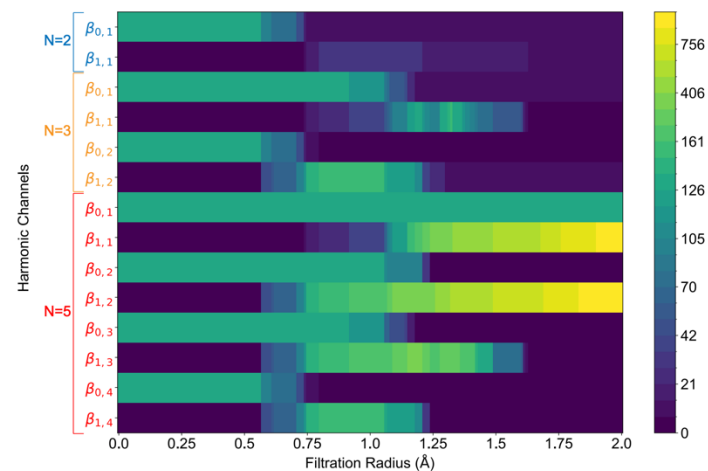
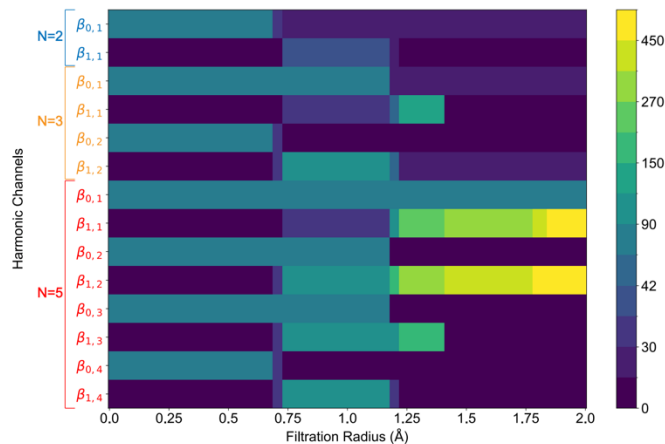
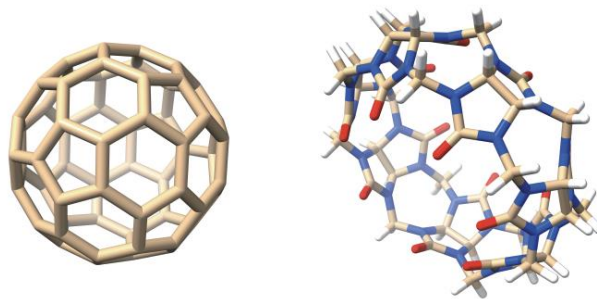
Born at time b_t never vanishes

Born at time b_s and vanishes at $b_s + c_s$

Persistent Mayer Homology example



The fullerene C60 (Left) and the cucurbit[7]uril CB7



Persistent Mayer Diagrams

r-Wasserstein Distance for Persistent Diagram

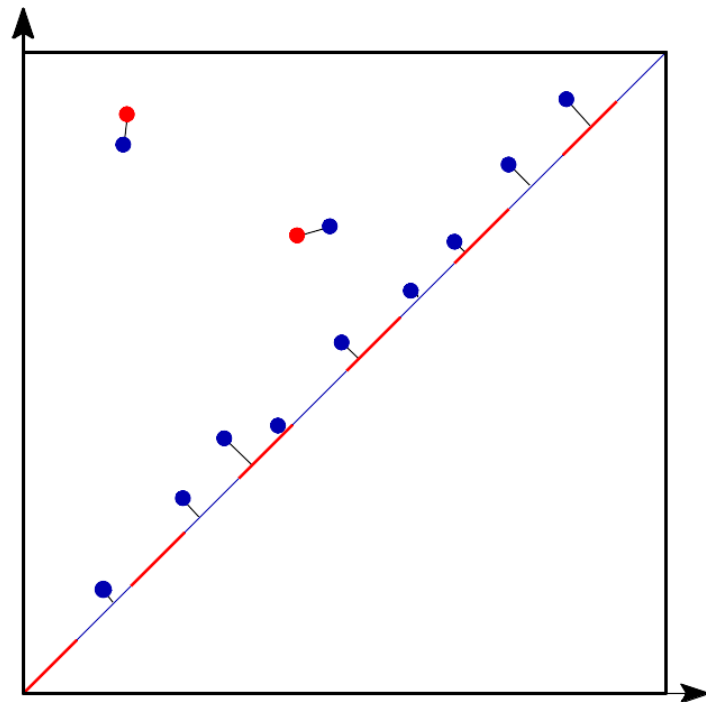
$$W_r(\mathcal{D}, \mathcal{D}') = \inf_{\gamma: \mathcal{D} \rightarrow \mathcal{D}'} \left(\sum_{x \in \mathcal{D}} \|x - \gamma(x)\|_s^r \right)^{1/r}$$

r-Wasserstein Distance for Persistent Mayer Diagram

$$W_r(\{\mathcal{D}_q\}_{1 \leq q \leq N-1}, \{\mathcal{D}'_q\}_{1 \leq q \leq N-1}) = \left(\sum_{q=1}^{N-1} W_r(\mathcal{D}_q, \mathcal{D}'_q)^r \right)^{1/r}$$

When $r = \infty$, Shen showed that PMD is stable under Bottleneck Distance

$$d_B(\{\mathcal{D}_q\}_{1 \leq q \leq N-1}, \{\mathcal{D}'_q\}_{1 \leq q \leq N-1}) = \sup_{1 \leq q \leq N-1} \inf_{\gamma: \mathcal{D}_q \rightarrow \mathcal{D}'_q} \sup_{x \in \mathcal{D}_q} |x - \gamma(x)|$$



Mayer Laplacian on N-chains

$$\langle \lambda\sigma, \mu\tau \rangle = \begin{cases} \lambda \cdot \bar{\mu}, & \sigma = \tau; \\ 0, & \text{otherwise.} \end{cases}$$

Mayer Laplacian $\Delta_{*,q} := (d^q)^* \circ d^q + d^{N-q} \circ (d^{N-q})^*.$

$$L_q = B^q (\overline{B}^q)^T + (\overline{B}^{N-q})^T B^{N-q}.$$

$$\Delta_{n,q} = (d_n)^* \circ \cdots \circ (d_{n-q+1})^* \circ d_{n-q+1} \circ \cdots \circ d_n + d_{n+1} \circ \cdots \circ d_{n+N-q} \circ (d_{n+N-q})^* \circ \cdots \circ (d_{n+1})^*.$$

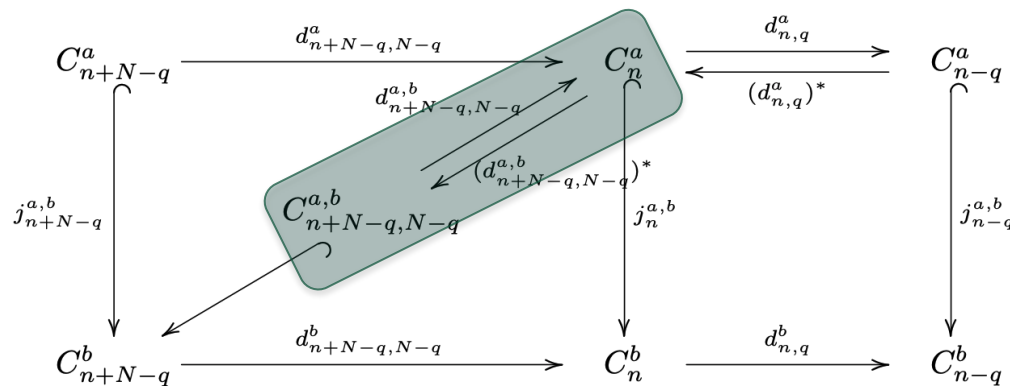
$$L_{n,q} = B_n \cdots B_{n-q+1} \overline{B_{n-q+1}}^T \cdots \overline{B_n}^T + \overline{B_{n+1}}^T \cdots \overline{B_{n+N-q}}^T B_{n+N-q} \cdots B_{n+1}.$$

For any n and $1 \leq q \leq N-1$, we have $\dim \ker \Delta_{n,q} = \beta_{n,q}$.

Persistent Mayer Laplacian

Lived from a to b in the filtration under q map

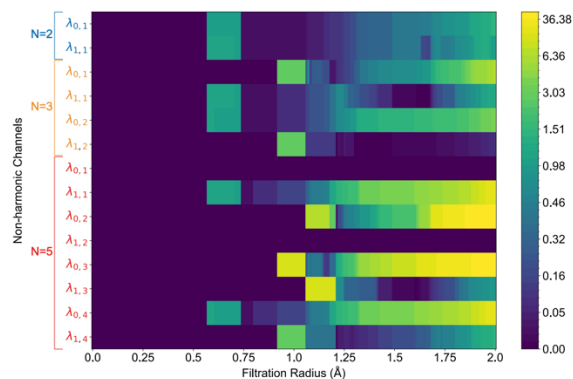
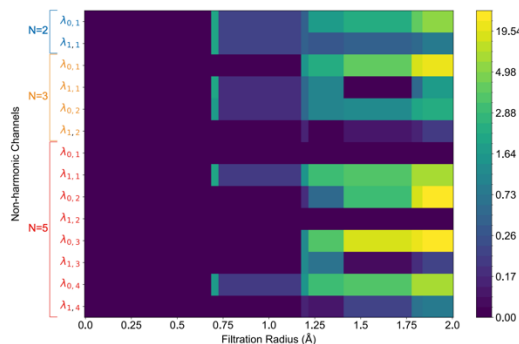
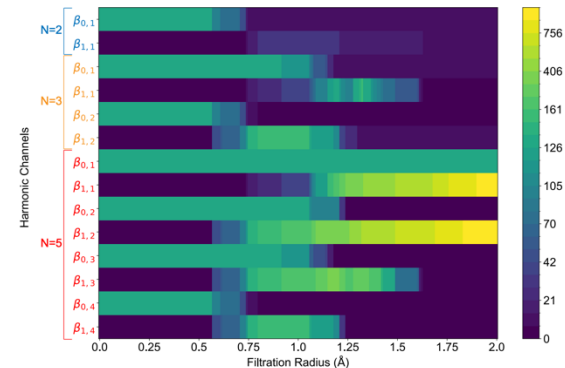
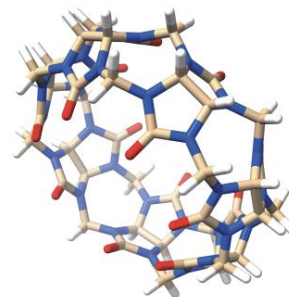
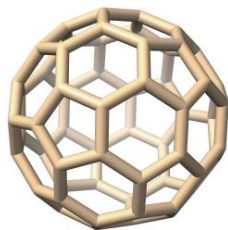
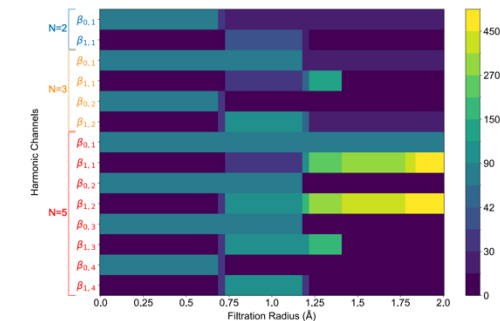
$$C_{n,q}^{a,b} = \{x \in C_n^b \mid d_{n,q}^b x \in C_{n-q}^a\}, \quad 1 \leq q \leq N-1.$$



$$\Delta_{n,q}^{a,b} := (d_{n,q}^a)^* \circ d_{n,q}^a + d_{n+N-q, N-q}^{a,b} \circ (d_{n+N-q, N-q}^{a,b})^*$$

$$\lambda_{n,q}^{a,b}(1), \lambda_{n,q}^{a,b}(2), \dots, \lambda_{n,q}^{a,b}(r),$$

The fullerene C60 (Left) and the cucurbit[7]uril CB7



PMH learning models for drug design

Mayer-homology learning prediction of protein-ligand binding affinities

Hongsong Feng, Li Shen, Jian Liu, Guo-Wei Wei

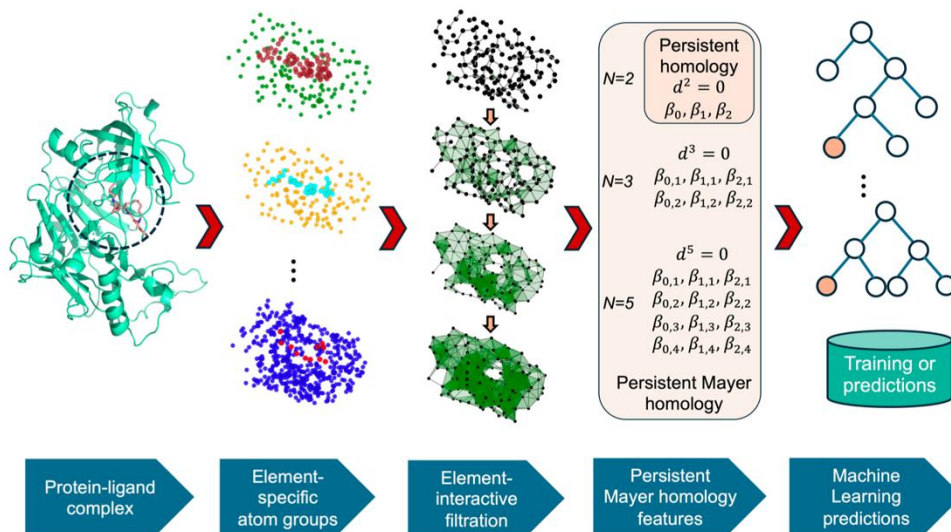


Figure 3: The illustration of persistent Mayer homology feature extraction for a protein-ligand complex (PDBID: 1A94) and the subsequent machine learning model development.

Protein Data Bank(PDB) bind dataset

The **PDBbind dataset** is a collection of **protein–ligand complexes** — each one being a **3D geometric configuration** of atoms.

Each complex :

$$X = \{(x_i, t_i, q_i)\}_{i=1}^M$$

where:

- $x_i \in \mathbb{R}^3$ are **atomic coordinates**,
- t_i denotes the **atom type** (e.g., C, N, O, S, ...),
- q_i represents **partial charges** or other local chemical features.

Each complex comes with an experimentally measured **binding free energy**, ΔG_{bind} , which quantifies **how strongly** the ligand binds to the protein..

The dataset as:

$$\{(X_j, \Delta G_j)\}_j^M$$

A **supervised learning dataset**

Protein Ligand Complex $\rightarrow$ feature vector

1) Vertices:

- Protein atoms : $S_P = \{C, N, O, S\}$
- Ligand atoms : $S_L = \{C, N, O, S, P, F, Cl, Br, I\}$
- 36 combination of P-L bind

2) Edges:

Interaction Regions :

- S_{X-Y}^c : all ligand atoms and all protein atoms **within distance c** of them

$$S_{X-Y}^c = \{a | a \in X, \min_{b \in Y} d(a, b) \leq c\} \cup \{b | b \in Y\}$$

Distance: The FRI function (Flexibility-Rigidity Index)

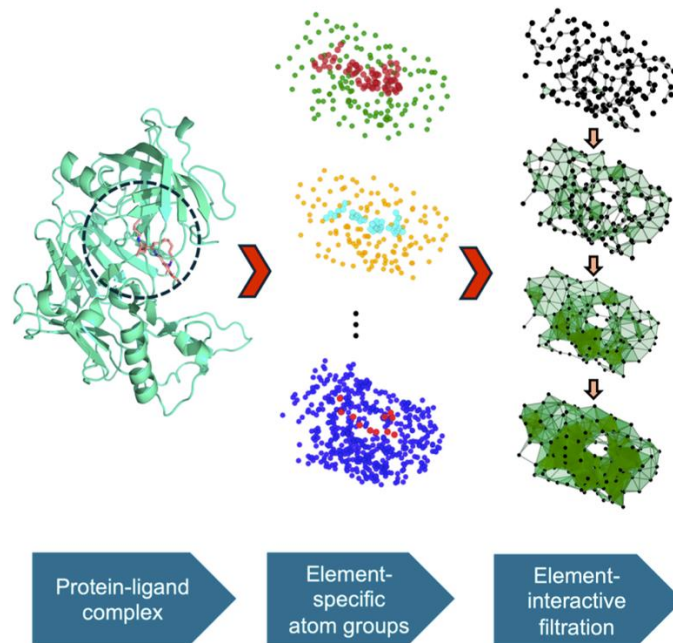
- The **FRI metric** replaces raw distances with a **smooth decaying function**:

$$d(i, j) = 1 - e^{-(r_{ij}/\eta_{ij})^\kappa}, \quad \eta_{ij} = \tau(r_i + r_j), \quad d_{Euc}(i, j) = r_{ij}$$

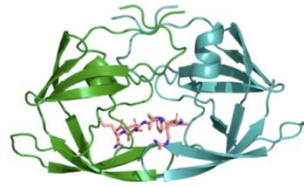
This function:

- r_i van der Waals radius- personal space of atom
- It reflects **interaction strength** rather than pure distance.

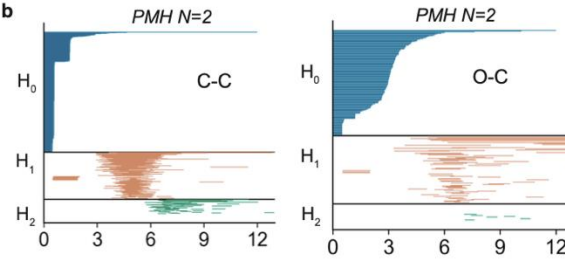
- Kelin Xia, Kristopher Opron, and Guo-Wei Wei. Multiscale multiphysics and multidomain models—flexibility and rigidity.



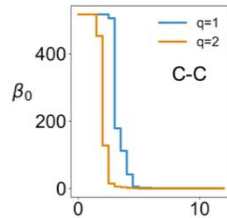
a Protein-ligand complex



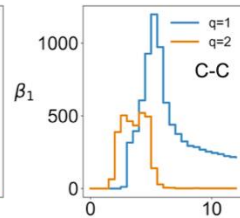
b



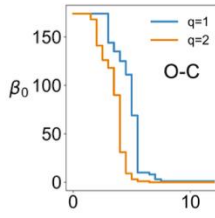
c *PMH N=3*



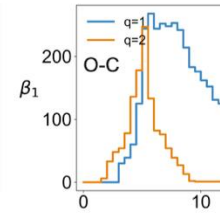
PMH N=3



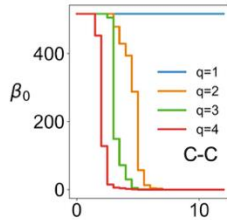
d *PMH N=3*



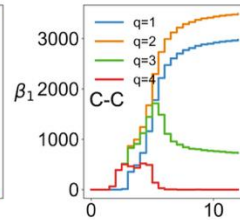
PMH N=3



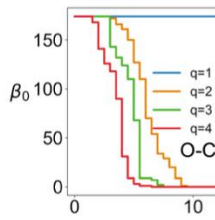
e *PMH N=5*



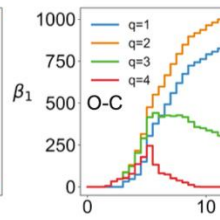
PMH N=5



f *PMH N=5*



PMH N=5

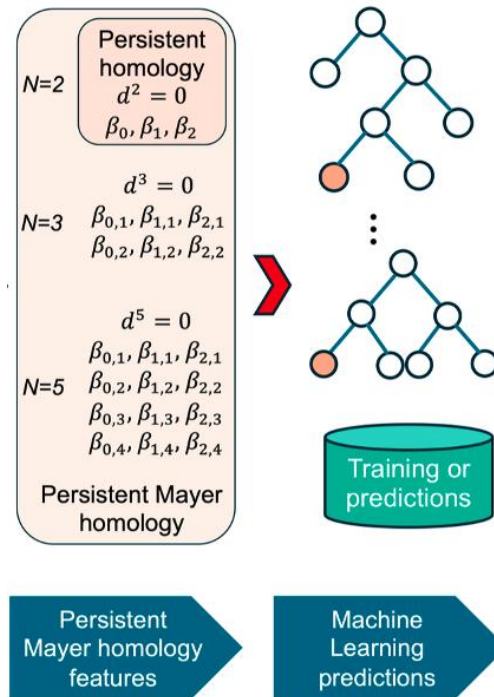


Persistent Mayer Homology features

Model-PMH2	I	$PMH2(P_{ep-el}^{12}, FRI^{agst}, VR)$ $ep \in S_P, el \in S_L$	Length sum of all Betti-0 bars.
	II	$PMH2(P_{all}^6, FRI, VR)$ $PMH2(P_{pro}^6, FRI, VR)$	Length sum and birth sum of Betti-0, Betti-1, and Betti-2 bars for protein, complex, as well as the sum differences between protein and complex.
	III	$PMH2(P_{ep-el}^{12}, EUC^{agst}, VR)$ $ep \in S_P, el \in S_L$	Counts of Betti-0 bars with 'death' values within each interval: [0, 2.5], [2.5, 3], [3, 3.5], [3.5, 4.5], [4.5, 6], [6, 12].
	IV	$PMH2(P_{all}^9, EUC, Alpha)$ $PMH2(P_{pro}^9, EUC, Alpha)$	Length sum of Betti-1 and Betti-2 bars with 'birth' values within each interval: [0, 2], [2, 3], [3, 4], [4, 5], [5, 6], [6, 9]. The sum differences between complex and protein are also considered.
Model-PMH5	V	$PMH5(P_{ep-el}^{12}, EUC, Alpha)$ $ep \in S_P \setminus \{C\}, el \in S_L \cup \{H\}$ $ep \in \{C\}, el \in S_L \cup \{H\}$	$\beta_{k,q}$ ($k=0,1, q=1,2,\dots,4$) over filtration parameter range from 0 to 10 with stepsize of 0.2. $\beta_{k,q}$ ($k=0,1, q=1,2,\dots,4$) over filtration parameter range from 0 to 8 with stepsize of 0.5.

No. of estimators	Max depth	Min. sample split	Learning rate
20000	7	5	0.002
Max features	Subsample size	Repetition	
Square root	0.8	20 times	

Table 3: Hyperparameters used to build gradient boosting regression models.



Final = the consensus of Model-PMH2 and Model-PMH5

$$y_{pred} = \frac{y_{PMH2} + y_{PMH5}}{2}$$


$$y_{pred} = \frac{y_{PMH2} + y_{PMH5} + y_{NLP}}{3}$$


Dataset	PMH	Transformer	PMH+Transformer
PDBbind-v2007	0.824(1.95)	0.795(2.006)	0.837(1.907)
PDBbind-v2013	0.787(2.036)	0.791(1.977)	0.807(1.982)
PDBbind-v2016	0.834(1.755)	0.836(1.716)	0.851(1.701)
Average	0.815 (1.914)	0.807 (1.9)	0.832 (1.863)

Table 4: Modeling performance of different strategies on the test sets of PDBbind-v2007, PDBbind-v2013 and PDBbind-v2016. Pearson correlation coefficient and root mean square error (unit, kcal/mol) are the two evaluation metrics.

